
Transient Refusal Dynamics: Detecting Late-Layer Escape in Instruction-Tuned Transformers

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Abstract

Understanding how refusal behavior emerges inside instruction-tuned language models is important to develop lightweight and interpretable safety diagnostics. We study refusal not as a static hidden-state classification problem, but as a trajectory in a residual-stream representation space. Using Qwen2.5-3B-Instruct, we construct a refusal-associated reference trajectory from a separate set of consistently refused prompts and measure how new prompts move relative to this trajectory across transformer layers. On adversarial prompts from JailbreakBench, we find that provisionally successful jailbreak outcomes exhibit a larger localized departure from the refusal trajectory in late layers, while overall layer-to-layer representation movement remains similar across successful and failed outcomes. A simple late-layer escape statistic achieves an AUROC of 0.832 on 57 non-ambiguous examples, with a 5,000-trial permutation test yielding $p \approx 2 \times 10^{-4}$. Matched-topic and framing controls show that near-perfect static hidden-state classification can be strongly confounded by prompt semantics, motivating the trajectory-based formulation. These results provide preliminary evidence that refusal-related behavior may be better characterized by localized internal dynamics than by global representational separation. We view the proposed escape statistic as a lightweight, model-internal diagnostic whose generality and causal relevance remain to be tested across models and with independent jailbreak evaluators.

1 Introduction

Instruction-tuned language models often refuse unsafe requests [2], yet adversarial prompts can sometimes bypass this behavior. A central mechanistic question is therefore not only whether successful and failed jailbreaks [11] have different internal representations, but *where* their internal computation begins to diverge. A natural first approach is to examine hidden states at individual transformer layers and test whether they linearly separate safety-related behaviors.

Recent work in representation engineering [14] has shown that safety mechanisms can be isolated and that refusal behavior is largely mediated by a single directional subspace within the residual stream [1]. Subsequent studies suggest a richer picture that involves multiple safety-relevant directions and interacting features [5, 6, 8]. However, static separation can be deceptively easy. Prompt topic, lexical cues, authorization language, or safety framing may all produce highly separable representations without identifying a mechanism specific to refusal.

This issue appeared directly in our initial experiments. Simple layerwise hidden-state classifiers achieved near-perfect separation between benign and safety-related prompts. The effect persisted under stronger controls involving matched topics and different safety framings, suggesting that static

representations strongly encode prompt semantics even when the resulting behavioral interpretation remains unclear.

We therefore study a different object, that is, the *trajectory* followed by the residual-stream representation across the transformer depth. Rather than asking whether two classes occupy different regions of activation space, we ask whether jailbreak-related outcomes exhibit a localized departure from a refusal-associated internal path.

Our main contribution is a lightweight trajectory statistic measuring layer-to-layer “escape” from an independently defined refusal reference. On Qwen2.5-3B-Instruct evaluated on adversarial Jailbreak-Bench prompts [15], this statistic identifies a localized late-layer difference between provisionally successful and failed outcomes. Of what follows, generic residual-stream motion remains similar across the two groups, suggesting that the effect is relative to the refusal trajectory rather than a global increase in representational change.

We highlight that our claims are deliberately limited. The behavioral labels are provisional, the layer window was identified exploratorily, and we do not establish causality. The purpose of this study is instead to isolate a concrete internal signature that can subsequently be independently judged, replicated, and intervened on.

2 Method

2.1 Residual-stream trajectories

We use Qwen2.5-3B-Instruct, an instruction-tuned member of the Qwen2.5 model family [9]. For each prompt x , we extract the hidden representation of the final prompt token at every hidden-state level, $h_0(x), h_1(x), \dots, h_L(x)$, with $L = 36$ and hidden dimension 2048.

Since the residual-stream norm grows strongly with depth, all states are normalized:

$$\hat{h}_\ell(x) = \frac{h_\ell(x)}{\|h_\ell(x)\|_2 + \varepsilon}.$$

Thus, the object of interest is the sequence $\hat{h}_0(x), \hat{h}_1(x), \dots, \hat{h}_L(x)$, rather than any single hidden vector in isolation.

2.2 Refusal reference trajectory

We construct a refusal reference from a separate set $\mathcal{R}$ of 30 matched unsafe prompts that Qwen2.5-3B-Instruct consistently refuses.

At each layer, we compute $\tilde{r}_\ell = \frac{1}{|\mathcal{R}|} \sum_{x \in \mathcal{R}} \hat{h}_\ell(x)$, and renormalize it as $r_\ell = \frac{\tilde{r}_\ell}{\|\tilde{r}_\ell\|_2 + \varepsilon}$.

For a new prompt x , its distance from the refusal trajectory at the layer ℓ is $d_\ell(x) = \|\hat{h}_\ell(x) - r_\ell\|_2$.

We define the local escape increment $e_\ell(x) = d_{\ell+1}(x) - d_\ell(x)$. Positive $e_\ell(x)$ means that the representation moves further away from the refusal-associated trajectory between consecutive layers.

2.3 Late-layer escape score

Exploratory analysis showed the clearest group difference in a late-layer region preceding the large final transition shared by both groups.

We therefore define $E(x) = \max_{\ell=22, \dots, 30} e_\ell(x)$. The interval 22–30 was selected after inspecting the trajectories and is therefore treated as an exploratory analysis window rather than a preregistered choice.

3 Experimental Design

3.1 Static representation controls

We first tested whether layerwise hidden states alone could separate safety-related behaviors. An initial synthetic set contained 60 prompts:- 20 benign prompts, 20 direct unsafe prompts, and 20 contextual safety prompts. The layerwise classification between benign and direct unsafe prompts was nearly perfect. To test whether this reflected refusal rather than semantic content, we built a second dataset of 30 matched topic triplets, for a total of 90 prompts. Each triplet contained a benign request, a direct unsafe request, and a contextual safety version concerning the same underlying topic.

Topic-held-out classification remained highly predictive. Direct, contextual, and unsafe safety prompts were also nearly perfectly separable. These results indicated that static hidden states encode topic and framing strongly enough to make classification accuracy difficult to interpret mechanistically. We therefore use these experiments as controls rather than as the primary result.

3.2 JailbreakBench evaluation

For trajectory analysis, we sampled 60 adversarial prompts from the JailbreakBench judge-comparison dataset [3]. JailbreakBench provides standardized adversarial prompts and evaluation tools for studying jailbreak attacks and defenses. For each prompt, we generated a Qwen2.5-3B-Instruct response and extracted the full final-prompt-token residual trajectory. The human labels included in the judge-comparison dataset apply to the stored benchmark responses rather than to the newly generated Qwen outputs. We therefore constructed provisional behavioral labels using: (i) explicit refusal-language detection and (ii) semantic similarity between the benchmark goal and the generated response. The responses were categorized as provisional failure, provisional success, or ambiguous. Three ambiguous examples were excluded, leaving 57 examples: 22 provisional failures and 35 provisional successes.

4 Results

4.1 Generic movement does not explain the effect

We first measured the total normalized layer-to-layer movement, $g_\ell(x) = \|\hat{h}_{\ell+1}(x) - \hat{h}_\ell(x)\|_2$. The resulting trajectories were broadly similar between explicit refusals and responses without explicit refusal language. Thus, provisional jailbreak success was not associated with a uniformly larger residual-stream movement.

4.2 Late-layer escape from refusal

The difference became clearer when the movement was measured relative to the separate refusal reference trajectory. Figure 1 shows the mean increment in layerwise escape for provisional successes and failures. The trajectories remain broadly similar across much of the model, but provisional successes exhibit larger localized positive excursions in the late-layer region.

The mean escape scores were $\bar{E}_{\text{failure}} = 0.0715$, $\bar{E}_{\text{success}} = 0.1066$. Using $E(x)$ alone to predict provisional outcome yields AUROC = 0.8325. A 5,000-trial label-permutation test gives $p \approx 2.0 \times 10^{-4}$, and the bootstrap 95% confidence interval for AUROC is [0.693, 0.943]. This supports a narrower interpretation than the original static-classification result, that is, the relevant difference appears as a localized departure relative to a refusal-associated trajectory rather than a globally larger change in hidden representation.

5 Discussion and Limitations

The main empirical observation is that a simple trajectory-relative statistic contains substantial information about the provisional jailbreak outcome. This complements prior work showing that refusal-related behavior can be associated with low-dimensional representational structure [1] and intervention directions [7]. Recent work has also examined jailbreak mechanisms through representation and circuit-level changes across models [4, 12]. Our focus differs in emphasizing the *layerwise*

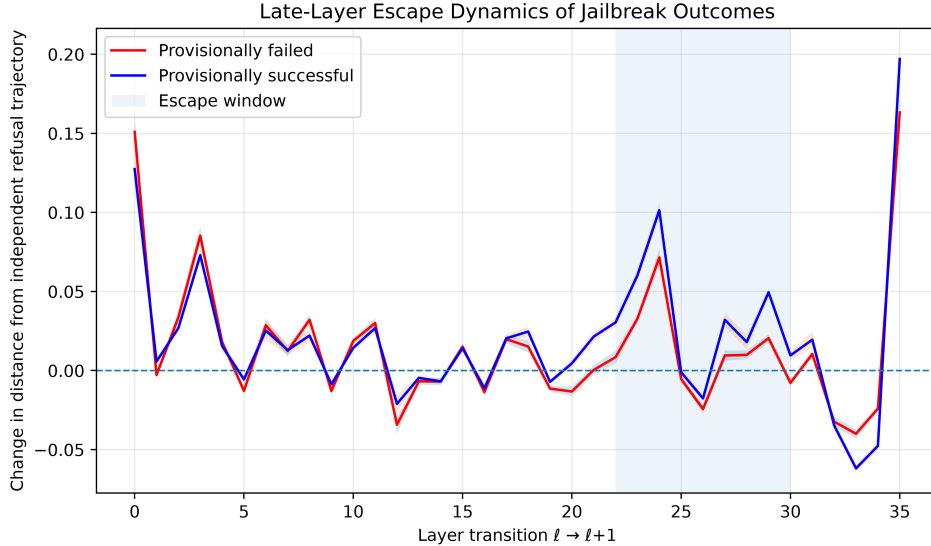


Figure 1: **Layerwise escape from an independently constructed refusal trajectory.** Provisionally successful outcomes exhibit a larger localized departure in the exploratory late-layer window (shaded), while overall layer-to-layer dynamics remain similar across groups.

path relative to a refusal reference, together with controls showing why static separability alone can be misleading.

Limitations. First, the final behavioral labels are heuristic. An independent established jailbreak evaluator should replace the provisional labeling procedure before treating the reported AUROC as definitive. Second, the study uses one model and a small sample. The same trajectory statistic should be frozen and tested on a second modern instruction-tuned model. Third, the layer window was selected after observing the data. The current result should therefore be regarded as exploratory. Fourth, we do not establish causality. Activation steering or activation patching [14, 10, 13] in the identified region would be a natural next test to see whether escape participates in the behavioral transition or simply correlates with it. Finally, we do not provide a broad benchmark against alternative jailbreak detectors, static refusal directions, cosine-similarity baselines, or equivalent trajectory statistics in other models. The present study instead focuses on establishing whether a localized refusal-relative departure remains visible after controlling for prompt topic, framing, residual norm growth, and generic layer-to-layer motion. A broader comparative evaluation is therefore left for future work.

6 Conclusion

Static hidden-state separability can provide numerically strong but mechanistically ambiguous evidence for safety-related behavior. By treating the residual stream as a layerwise trajectory, we identify a more localized candidate signature, that is, late-layer escape from an independently constructed refusal reference. On Qwen2.5-3B-Instruct, this statistic separates provisionally successful and failed JailbreakBench outcomes with an AUROC of 0.8325, while generic layer-to-layer movement remains broadly similar between the two groups. These findings suggest a representation-level hypothesis for mechanistic interpretability, i.e. jailbreak success may involve a localized departure from a refusal-associated computational path rather than a globally distinct hidden state. The present evidence is correlational and exploratory. Establishing a mechanism will require testing whether interventions in the identified late-layer region—for example through activation patching or steering—can causally alter the eventual refusal behavior.

7 Broader Impact

Understanding how safety-related behavior develops within language models may support more interpretable and reliable safety mechanisms. In particular, identifying where refusal-related representations begin to diverge could help researchers localize safety-relevant computation, evaluate whether observed behavior reflects stable internal structure, and design more targeted tests of model safeguards. The trajectory-based analysis considered here may, therefore, contribute to the mechanistic understanding of refusal and to the development of more transparent model-safety evaluations. At the same time, mechanistic information about refusal behavior is inherently dual-use. A better understanding of where and how refusal-related computation changes could potentially be used not only to strengthen safeguards but also to design interventions intended to weaken or circumvent them. Our study does not develop a new jailbreak attack, release modified model weights, or provide a procedure for disabling safety mechanisms. We report aggregate representation-level measurements and emphasize their use to understand and evaluate safety behavior. More generally, any future causal intervention based on the identified late-layer region should be evaluated with both defensive utility and potential misuse in mind.

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A Reproducibility Details

Model and representation extraction. All experiments use Qwen2.5-3B-Instruct. For each input prompt, we record the hidden representation of the final prompt token at every hidden-state level, giving a trajectory $h_0(x), h_1(x), \dots, h_{36}(x)$, with hidden dimension 2048. Because residual-stream norms grow substantially with depth, each representation is normalized as

$$\hat{h}_\ell(x) = \frac{h_\ell(x)}{\|h_\ell(x)\|_2 + \epsilon}.$$

The resulting normalized trajectories, rather than individual hidden states, form the input to the trajectory analysis.

Refusal reference and escape statistic. The refusal reference is constructed independently from 30 matched unsafe prompts for which Qwen2.5-3B-Instruct consistently produces refusal behavior. At each layer,

$$\tilde{r}_\ell = \frac{1}{30} \sum_{x \in \mathcal{R}} \hat{h}_\ell(x), \quad r_\ell = \frac{\tilde{r}_\ell}{\|\tilde{r}_\ell\|_2 + \epsilon}.$$

For an evaluation prompt x , we compute

$$d_\ell(x) = \|\hat{h}_\ell(x) - r_\ell\|_2, \quad e_\ell(x) = d_{\ell+1}(x) - d_\ell(x),$$

and define the late-layer escape statistic by

$$E(x) = \max_{\ell=22, \dots, 30} e_\ell(x).$$

The interval 22–30 was selected after exploratory inspection of the trajectories and was not preregistered.

Statistical analysis. The primary evaluation uses the scalar escape statistic $E(x)$ to discriminate between provisional successful and failed outcomes. The resulting AUROC is 0.8325. Statistical significance is assessed using 5,000 random permutations of the provisional outcome labels, yielding $p \simeq 2.0 \times 10^{-4}$. A nonparametric bootstrap over the 57 retained examples gives a 95% confidence interval of [0.693, 0.943] for AUROC. The corresponding mean escape scores are 0.0715 for provisional failures and 0.1066 for provisional successes.

Generic-motion control. As a control for overall representational movement, we additionally compute $g_\ell(x) = \|\hat{h}_{\ell+1}(x) - \hat{h}_\ell(x)\|_2$. The resulting trajectories are broadly similar between provisional successful and failed outcomes, distinguishing the refusal-relative escape effect from a uniform increase in residual-stream movement.

Software and compute environment. Experiments were executed on an Apple Silicon system running macOS on an ARM64 architecture with 18 CPU cores and Metal Performance Shaders (MPS) available for PyTorch acceleration. The software environment used Python 3.12.13, PyTorch 2.13.0, NumPy 2.5.2, Transformers 5.16.1, Datasets 5.0.1, Accelerate 1.14.0, Tokenizers 0.23.1, Sentence-Transformers 6.0.0, scikit-learn 1.9.0, SciPy 1.18.1, and pandas 2.3.3. CUDA was not used.

Existing assets. We use the publicly released Qwen2.5-3B-Instruct checkpoint [9] and the JailbreakBench benchmark [3]. The experiments use these assets under their respective publicly stated licenses and terms of use; no model weights or benchmark data are redistributed as part of this work.

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